

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461124.8416 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1719 +/- 0.0092
Transit depth (Rp/Rs)^2	2.96 +/- 0.32 [%]
Orbital Inclination (inc)	89.94 +/- 0.8
Airmass coefficient 1 (a1)	0.911 +/- 0.016
Airmass coefficient 2 (a2)	0.088 +/- 0.078
Scatter in the residuals of the lightcurve fit is	1.73 %
Best Comparison Star	#2 - [153, 243]
Optimal Aperture	4.18
Optimal Annulus	10.35
Transit Duration (day)	0.1005 +/- 0.002

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1719 ± 0.0092 vs 0.1406 (deviation 3.4σ)
Duration fit vs published	0.1005 d vs 0.0975 d (deviation 1.5σ)
Mid-time O–C	-9.1 min
Depth SNR	18.7
Residual scatter	1.73 %

Light curve

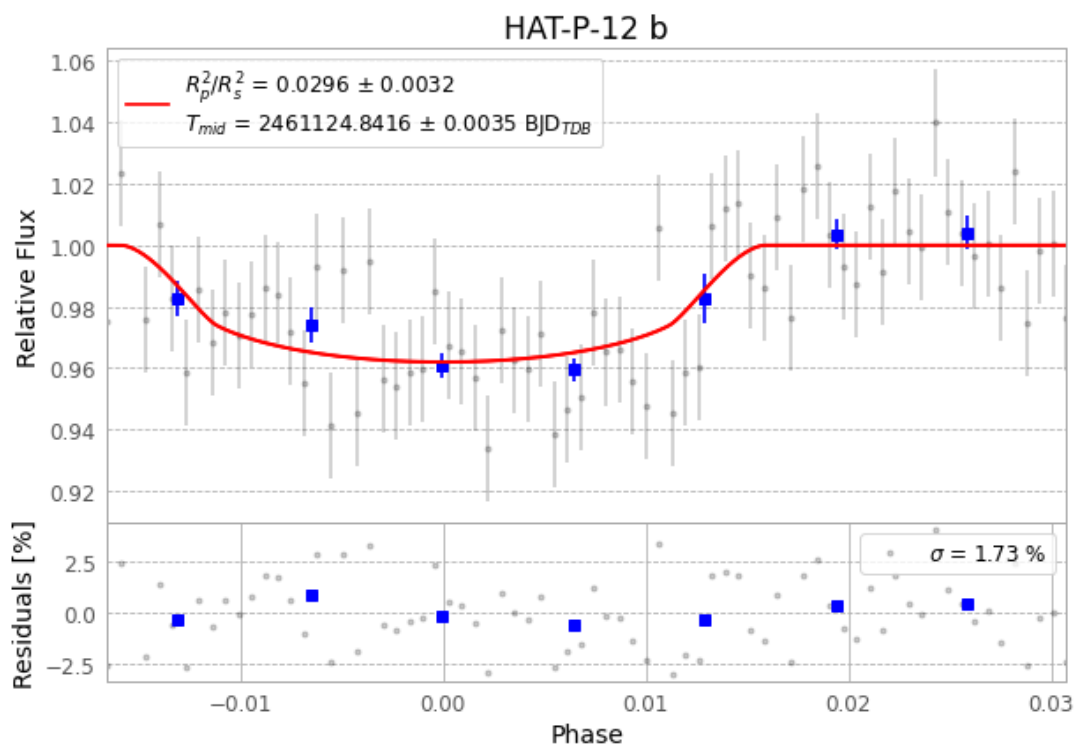


Figure 1 — FinalLightCurve_HAT-P-12 b_2026-03-24.png (SHA-256 20396e3292f28304...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [301, 202], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6f5de654635ad23e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

74 FITS frames (49 MB), HATP-12260325064816.FITS → HATP-12260325102715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-260325.log (e5b33b04eb13...), FinalLightCurve_HAT-P-12 b_2026-03-24.png (20396e3292f2...), FinalLightCurve_HAT-P-12 b_2026-03-24.csv (283e43e6b2ee...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 19:20Z.